

CLASS - XI PROJECT

Program

```
import java.io.*;
import java.util.Scanner;

public class Movie
{
    int choice, noOfAvailableSeats;
    Scanner sc;
    public static void main()
    {
        Movie obj = new Movie();
        do
        {
            obj.displayMenu();
            obj.getChoice();
            switch(obj.choice)
            {
                case 1: obj.bookTicket();
                        break;
                case 2: obj.cancelTicket();
                        break;
                case 3: obj.showAvailable(true);
                        break;
                case 4: obj.resetSeats();
                        break;
                case 5: System.out.println("\n*****Program Terminated*****");
                        System.exit(0);
                default: System.out.println("Invalid Choice!! Try Again!!");
            }
        }while(obj.choice != 5);

    }

    void getChoice()
    {
        System.out.print("\n\nPlease enter your choice: ");
        sc = new Scanner (System.in);
        choice = sc.nextInt();
        sc.close();
    }

    void displayMenu()
    {
        System.out.println("\n\n\n\n---Movie Ticket Booker---");
        System.out.println("-----");
        System.out.println("|    1.Book a ticket    |");
        System.out.println("|    2.Cancel a ticket  |");
        System.out.println("|    3.Check Availability |");
        System.out.println("|    4.Reset Seats      |"); //cancel all tickets
        System.out.println("|    5.Exit             |");
        System.out.println("-----");
    }

    void showAvailable(boolean printSeats)
```

```

{

    try
    {
        File myObj = new File("d:\\movie.txt");
        Scanner myReader = new Scanner(myObj);
        while (myReader.hasNextInt())
        {
            String data = myReader.nextLine();
            noOfAvailableSeats = Integer.valueOf(data);
            System.out.println("\nTotal Seats Available: " + noOfAvailableSeats);
        }
        myReader.close();
        if(printSeats)
            printSeats();
    }
    catch (FileNotFoundException e)
    {
        System.out.println("An error occurred.");
        e.printStackTrace();
    }
}

void printSeats()
{
    int count=0;
    System.out.println();
    for(int i=1;i<=noOfAvailableSeats; i++)
    {
        System.out.print("\u2b1b\t");
        count++;
        if(count == 10)
        {
            System.out.println("\n");
            count=0;
        }
    }
}

void cancelTicket()
{
    showAvailable(true); //read/update number of seats available in noOfAvailableSeats
variable
    sc = new Scanner(System.in);
    System.out.print("\n\nEnter how many tickets to cancel: ");
    int not = sc.nextInt(); //no of tickets need to be cancelled
    sc.close();

    if(not + noOfAvailableSeats > 100)
    {
        System.out.println("Tickets not cancelled. Maximum Tickets that can be
cancelled is: " + (100 - noOfAvailableSeats));
    }
    else
    {
        noOfAvailableSeats += not;
        try
        {
            FileWriter myWriter = new FileWriter("d:\\movie.txt");
            myWriter.write(noOfAvailableSeats+"");

```

```

        myWriter.close();
        System.out.println(not + " tickets successfully cancelled!!\n\n");
        printSeats();
    }
    catch (IOException e)
    {
        System.out.println("An error occurred.");
        e.printStackTrace();
    }
}

void bookTicket()
{
    showAvailable(true); //read/update number of seats available in noOfAvailableSeats
variable
    if(noOfAvailableSeats == 0)
    {
        System.out.println("\nAll tickets booked out. Sorry!!\n");
        return;
    }

    sc = new Scanner(System.in);
    System.out.print("\n\nEnter how many tickets: ");
    int not = sc.nextInt(); //no of tickets need to be booked
    sc.close();

    if(not <= noOfAvailableSeats) //no of tickets needed should be less than available
seats
    {
        noOfAvailableSeats = noOfAvailableSeats - not;
        try
        {
            FileWriter myWriter = new FileWriter("d:\\movie.txt");
            myWriter.write(noOfAvailableSeats+"");
            myWriter.close();
            System.out.println(not + " tickets successfully booked!! Enjoy the
movie!!\n\n");
            printSeats();
        }
        catch (IOException e)
        {
            System.out.println("An error occurred.");
            e.printStackTrace();
        }
    }
    else
    {
        System.out.println("\nNo. of tickets should be less than " +
noOfAvailableSeats);
        printSeats();
    }
}

void resetSeats()
{
    try
    {
        FileWriter myWriter = new FileWriter("d:\\movie.txt");
        myWriter.write("100");
    }
}

```


---Movie Ticket Booker---

```
-----  
|  1.Book a ticket  |  
|  2.Cancel a ticket |  
|  3.Check Availability |  
|  4.Reset Seats    |  
|  5.Exit           |  
-----
```

Please enter your choice: 2

Total Seats Available: 17

```
■      ■      ■      ■      ■      ■      ■      ■      ■      ■  
■      ■      ■      ■      ■      ■      ■
```

Enter how many tickets to cancel: 2

2 tickets successfully cancelled!!

```
■      ■      ■      ■      ■      ■      ■      ■      ■      ■  
■      ■      ■      ■      ■      ■      ■      ■      ■
```

OUTPUT 3

---Movie Ticket Booker---

```
-----  
|  1.Book a ticket  |  
|  2.Cancel a ticket |  
|  3.Check Availability |  
|  4.Reset Seats    |  
|  5.Exit           |  
-----
```

Please enter your choice: 3

Total Seats Available: 19

```
■      ■      ■      ■      ■      ■      ■      ■      ■      ■  
■      ■      ■      ■      ■      ■      ■      ■      ■
```

OUTPUT 4

---Movie Ticket Booker---

| 1.Book a ticket |
| 2.Cancel a ticket |
| 3.Check Availability |
| 4.Reset Seats |
5.Exit

Please enter your choice: 4
Reset complete!